

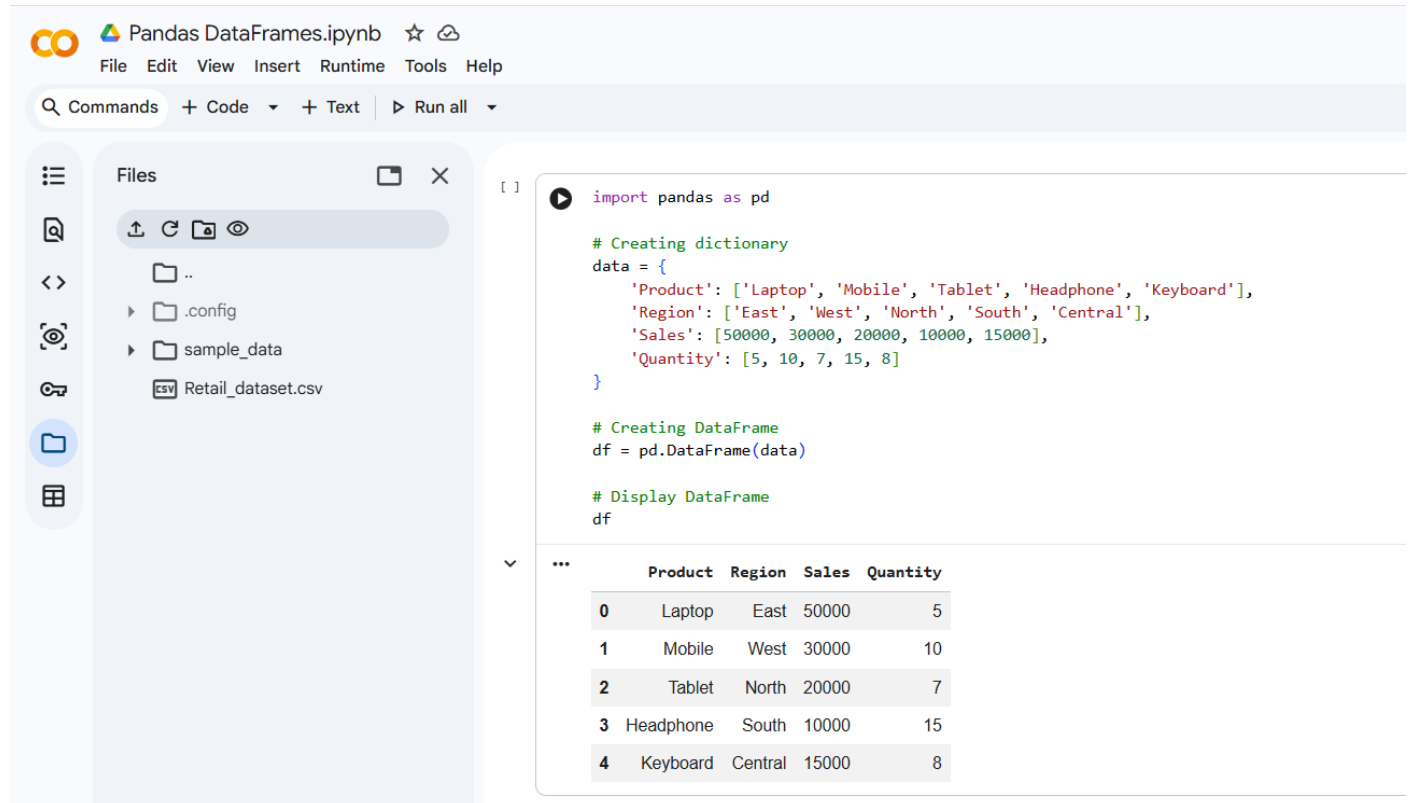
Retail Dataset

Q1. Creating a DataFrame

Create a Pandas DataFrame using a Python dictionary with the following columns:

- Product
- Region
- Sales
- Quantity

Ans:-



The screenshot shows a Jupyter Notebook interface with the title "Pandas DataFrames.ipynb". The left sidebar displays a file explorer with a folder named "sample_data" containing a file "Retail_dataset.csv". The main code cell contains the following Python code:

```
import pandas as pd

# Creating dictionary
data = {
    'Product': ['Laptop', 'Mobile', 'Tablet', 'Headphone', 'Keyboard'],
    'Region': ['East', 'West', 'North', 'South', 'Central'],
    'Sales': [50000, 30000, 20000, 10000, 15000],
    'Quantity': [5, 10, 7, 15, 8]
}

# Creating DataFrame
df = pd.DataFrame(data)

# Display DataFrame
df
```

The output of the code is a Pandas DataFrame with 5 rows and 5 columns. The columns are Product, Region, Sales, and Quantity. The rows are indexed from 0 to 4.

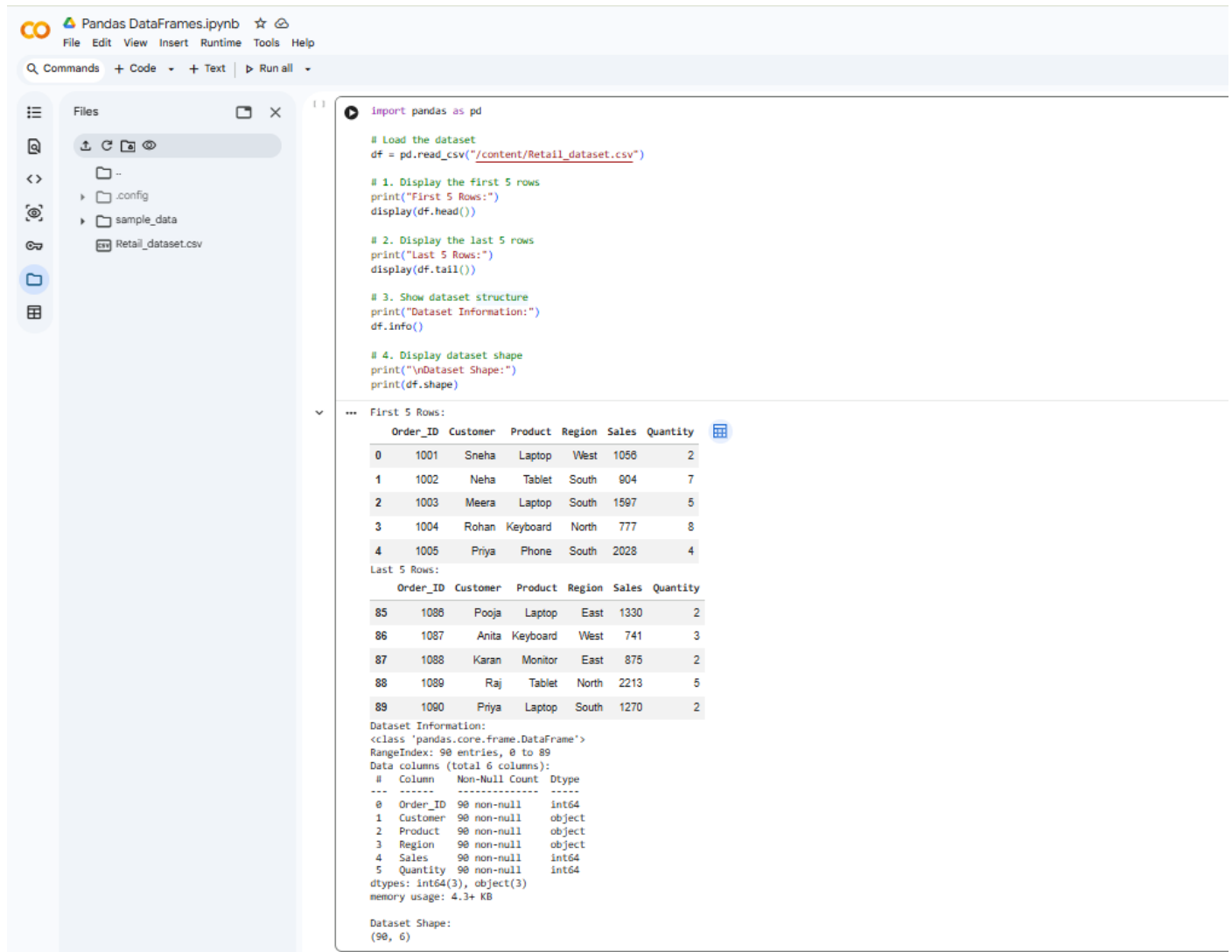
	Product	Region	Sales	Quantity
0	Laptop	East	50000	5
1	Mobile	West	30000	10
2	Tablet	North	20000	7
3	Headphone	South	10000	15
4	Keyboard	Central	15000	8

Q2. Data Inspection

Perform the following inspection tasks:

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows.
3. Show the structure of the dataset using info().
4. Display the shape of the dataset.

Ans:-



The screenshot shows a Jupyter Notebook titled "Pandas DataFrames.ipynb" with a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar (Commands, Code, Text, Run all). The left sidebar shows a file explorer with a folder named "sample_data" containing a file named "Retail_dataset.csv". The main area displays the following code:

```
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# 1. Display the first 5 rows
print("First 5 Rows:")
display(df.head())

# 2. Display the last 5 rows
print("Last 5 Rows:")
display(df.tail())

# 3. Show dataset structure
print("Dataset Information:")
df.info()

# 4. Display dataset shape
print("\nDataset Shape:")
print(df.shape)
```

The output of the code is displayed below the code cells:

First 5 Rows:

Order_ID	Customer	Product	Region	Sales	Quantity	
0	1001	Sneha	Laptop	West	1058	2
1	1002	Neha	Tablet	South	904	7
2	1003	Meera	Laptop	South	1597	5
3	1004	Rohan	Keyboard	North	777	8
4	1005	Priya	Phone	South	2028	4

Last 5 Rows:

Order_ID	Customer	Product	Region	Sales	Quantity	
85	1086	Pooja	Laptop	East	1330	2
86	1087	Anita	Keyboard	West	741	3
87	1088	Karan	Monitor	East	875	2
88	1089	Raj	Tablet	North	2213	5
89	1090	Priya	Laptop	South	1270	2

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 90 entries, 0 to 89
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Order_ID    90 non-null    int64
 1   Customer    90 non-null    object
 2   Product     90 non-null    object
 3   Region      90 non-null    object
 4   Sales       90 non-null    int64
 5   Quantity    90 non-null    int64
dtypes: int64(3), object(3)
memory usage: 4.3+ KB
```

Dataset Shape:

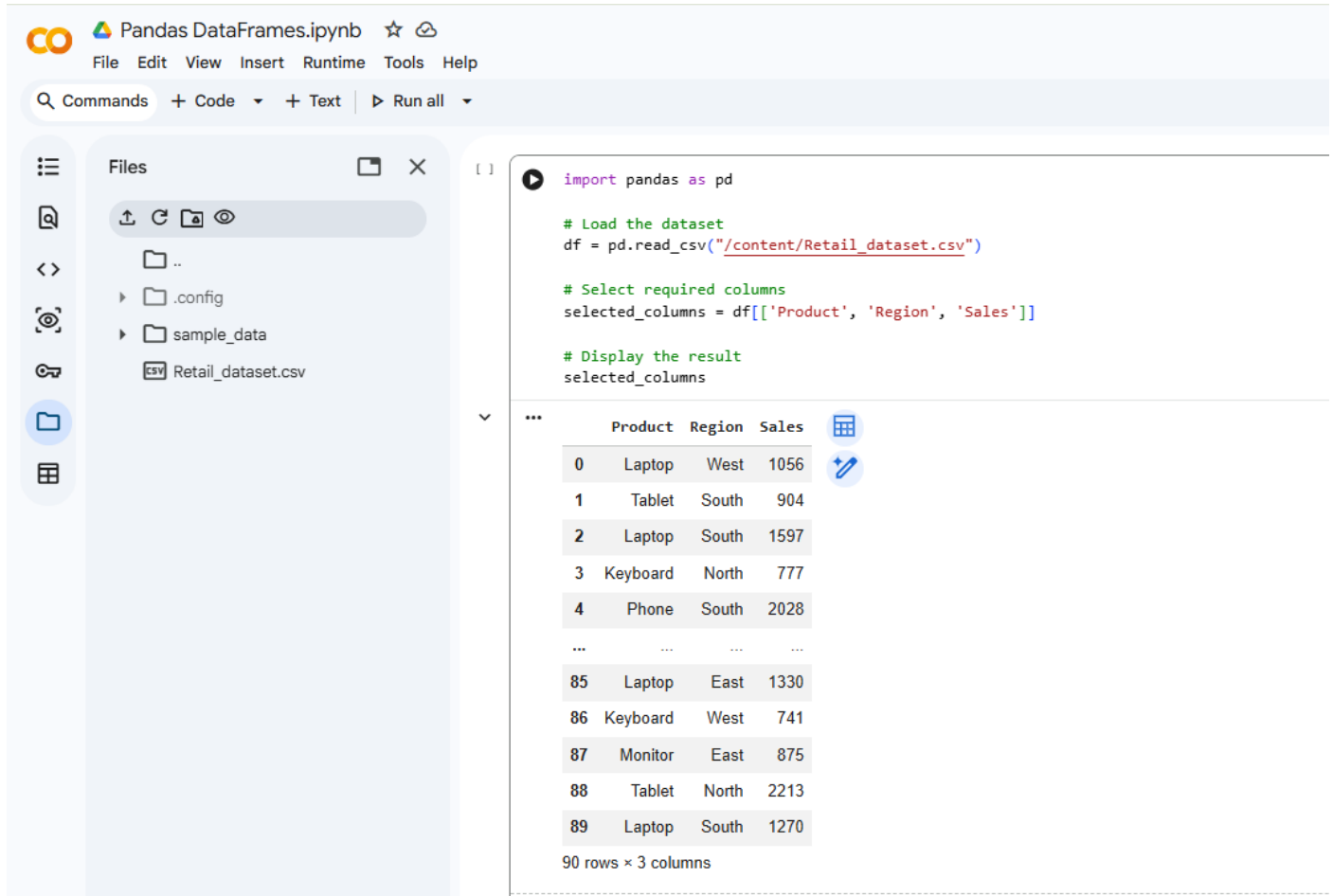
```
(90, 6)
```

Q3. Column Selection

Select and display only the following columns:

- Product
- Region
- Sales

Ans:-



The screenshot shows a Jupyter Notebook interface with the following components:

- Files Panel:** Displays a file explorer with a folder named 'sample_data' containing a file 'Retail_dataset.csv'.
- Code Cell:** Contains the following Python code:

```
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# Select required columns
selected_columns = df[['Product', 'Region', 'Sales']]

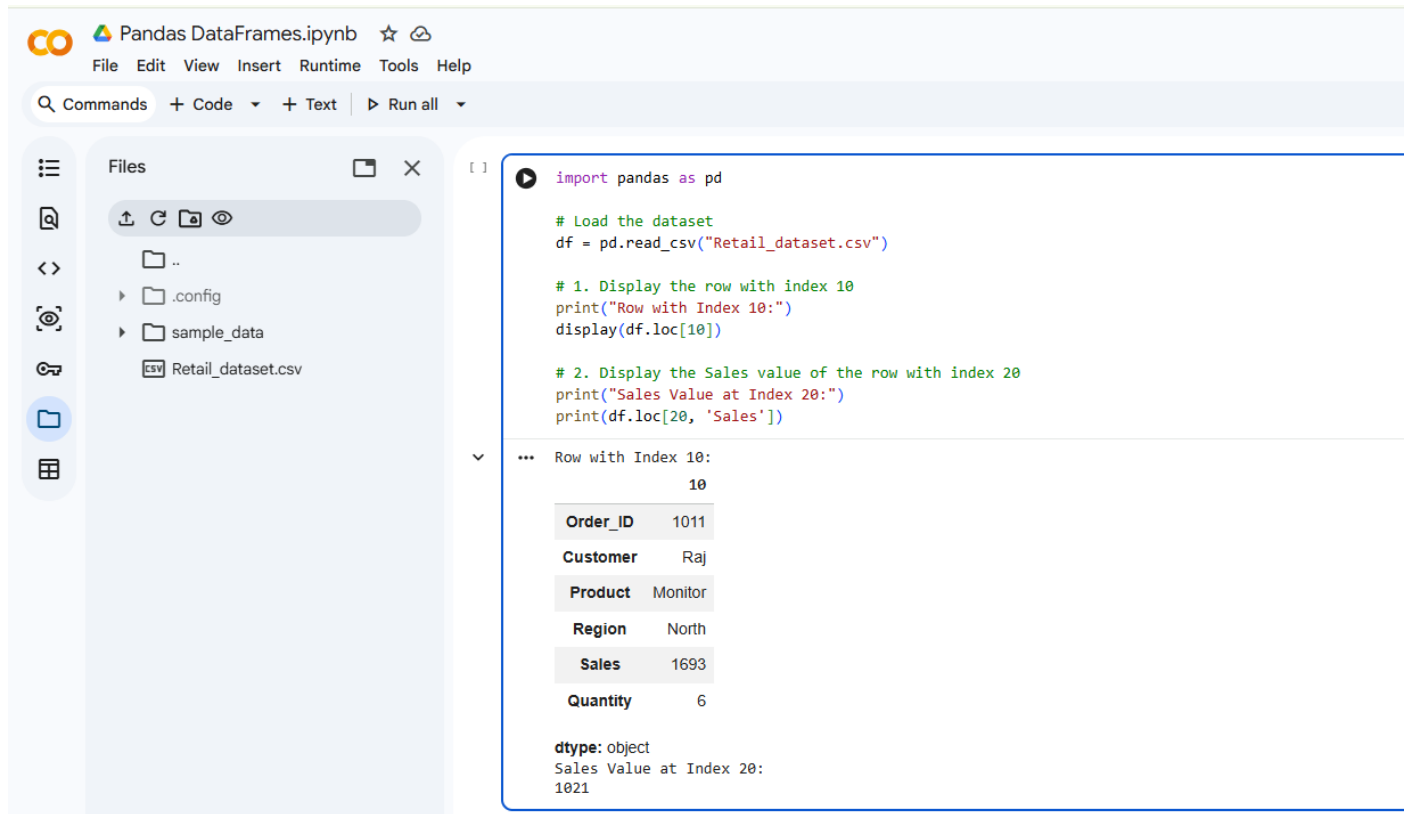
# Display the result
selected_columns
```
- Output:** A table with 90 rows and 3 columns (Product, Region, Sales). The first few rows are:

	Product	Region	Sales
0	Laptop	West	1056
1	Tablet	South	904
2	Laptop	South	1597
3	Keyboard	North	777
4	Phone	South	2028
...	...	...	...
85	Laptop	East	1330
86	Keyboard	West	741
87	Monitor	East	875
88	Tablet	North	2213
89	Laptop	South	1270

Q4. Indexing using .loc, perform the following:

1. Display the row with index 10.
2. Display the Sales value of the row with index 20

Ans:-



The screenshot shows a Jupyter Notebook interface with the following code and output:

```
import pandas as pd

# Load the dataset
df = pd.read_csv("Retail_dataset.csv")

# 1. Display the row with index 10
print("Row with Index 10:")
display(df.loc[10])

# 2. Display the Sales value of the row with index 20
print("Sales Value at Index 20:")
print(df.loc[20, 'Sales'])
```

The output for the first command shows the row with index 10:

	10
Order_ID	1011
Customer	Raj
Product	Monitor
Region	North
Sales	1693
Quantity	6

The output for the second command shows the Sales value at index 20:

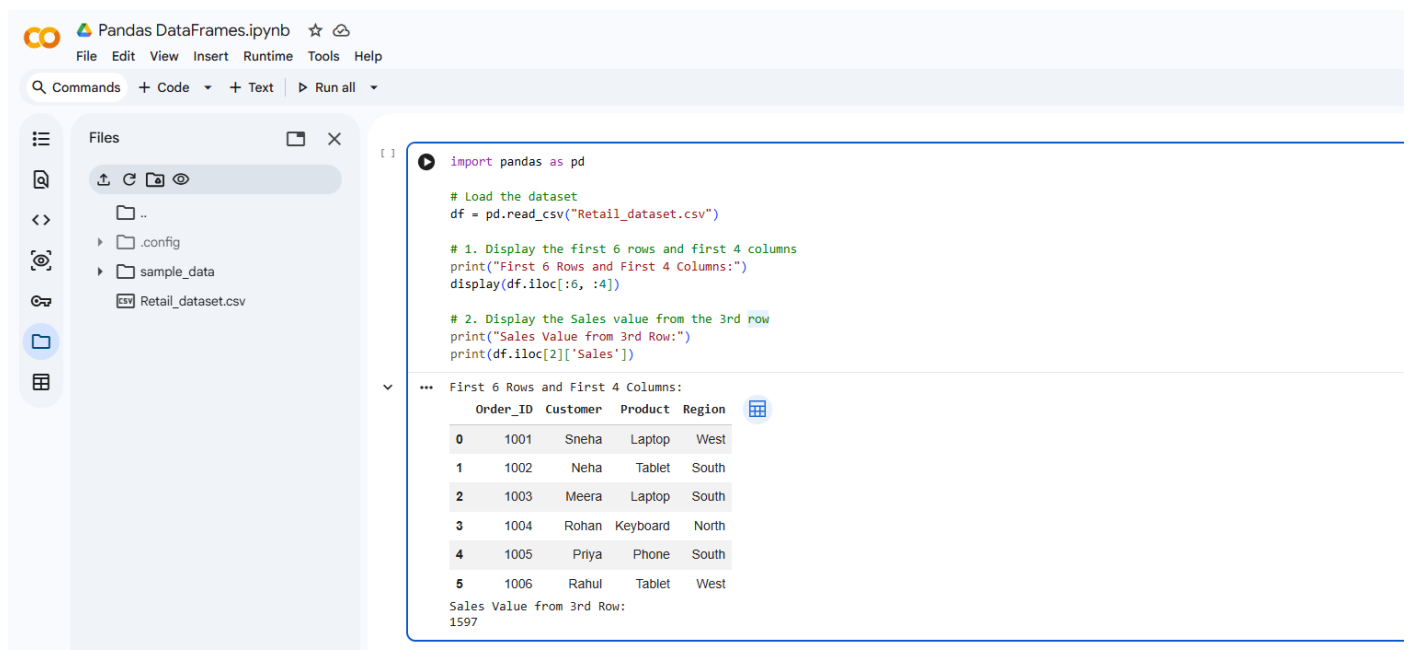
```
dtype: object
Sales Value at Index 20:
1021
```

Q5. Indexing using .iloc

Using .iloc, display:

1. The first 6 rows and first 4 columns.
2. The Sales value from the 3rd row.

Ans:-



The screenshot shows a Jupyter Notebook interface with the following code and output:

```
import pandas as pd

# Load the dataset
df = pd.read_csv("Retail_dataset.csv")

# 1. Display the first 6 rows and first 4 columns
print("First 6 Rows and First 4 Columns:")
display(df.iloc[:6, :4])

# 2. Display the Sales value from the 3rd row
print("Sales Value from 3rd Row:")
print(df.iloc[2]['Sales'])
```

The output for the first command shows the first 6 rows and first 4 columns:

	Order_ID	Customer	Product	Region
0	1001	Sneha	Laptop	West
1	1002	Neha	Tablet	South
2	1003	Meera	Laptop	South
3	1004	Rohan	Keyboard	North
4	1005	Priya	Phone	South
5	1006	Rahul	Tablet	West

The output for the second command shows the Sales value from the 3rd row:

```
Sales Value from 3rd Row:
1597
```

Display:

1. The first 10 rows of the dataset.
2. Rows 15 to 25.

Pandas DataFrames.ipynb

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Files

..
.config
sample_data
Retail_dataset.csv

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```
[ ] import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# 1. Display the first 10 rows
print("First 10 Rows:")
display(df[:10])

# 2. Display rows 15 to 25
print("Rows 15 to 25:")
display(df[15:26])
```

... First 10 Rows:

	Order_ID	Customer	Product	Region	Sales	Quantity
0	1001	Sneha	Laptop	West	1056	2
1	1002	Neha	Tablet	South	904	7
2	1003	Meera	Laptop	South	1597	5
3	1004	Rohan	Keyboard	North	777	8
4	1005	Priya	Phone	South	2028	4
5	1006	Rahul	Tablet	West	638	2
6	1007	Amit	Monitor	South	1004	7
7	1008	Sneha	Monitor	East	1269	8
8	1009	Priya	Phone	North	827	8
9	1010	Amit	Phone	West	1998	4

Rows 15 to 25:

	Order_ID	Customer	Product	Region	Sales	Quantity
15	1016	Karan	Monitor	West	796	3
16	1017	Neha	Laptop	East	2015	2
17	1018	Anita	Phone	North	1637	3

The screenshot shows a Jupyter Notebook interface. At the top, the title bar reads "Pandas DataFrames.ipynb". Below it is a menu bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". A toolbar contains "Commands", "+ Code", "+ Text", and "Run all". On the left, a "Files" sidebar shows a file explorer with a folder named "sample_data" containing a file named "Retail_dataset.csv". The main area displays a Pandas DataFrame with 7 columns and 10 rows (indices 18 to 25). The columns are: index, id, name, device, location, sales, and rating. The data is as follows:

	id	name	device	location	sales	rating
18	1019	Amit	Phone	East	748	7
19	1020	Neha	Laptop	West	1503	6
20	1021	Priya	Laptop	North	1021	4
21	1022	Raj	Tablet	South	886	7
22	1023	Rahul	Laptop	South	2267	3
23	1024	Karan	Monitor	East	814	7
24	1025	Meera	Monitor	West	1926	6
25	1026	Anita	Tablet	East	839	5

Q7. Filtering Data

Filter the dataset to display:

1. Rows where Region = "East".
2. Rows where Sales > 1500.

Ans:-

The screenshot shows a Jupyter Notebook interface with the file 'Retail_dataset.csv' in the left sidebar. The code cell contains the following Python code:

```
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# 1. Rows where Region = "East"
print("Rows where Region = East:")
display(df[df['Region'] == 'East'])

# 2. Rows where Sales > 1500
print(["Rows where Sales > 1500:"])
display(df[df['Sales'] > 1500])
```

The output of the first code block is displayed as a table:

Order_ID	Customer	Product	Region	Sales	Quantity	
7	1008	Sneha	Monitor	East	1269	8
11	1012	Rahul	Laptop	East	1022	5
16	1017	Neha	Laptop	East	2015	2
18	1019	Amit	Phone	East	748	7
23	1024	Karan	Monitor	East	814	7
25	1026	Anita	Tablet	East	839	5
31	1032	Amit	Laptop	East	2226	6
34	1035	Amit	Tablet	East	2104	3
35	1036	Rohan	Phone	East	2136	8
36	1037	Raj	Phone	East	1650	4
37	1038	Amit	Monitor	East	1567	8
38	1039	Meera	Keyboard	East	2455	3
44	1045	Priya	Phone	East	1875	4
47	1048	Rohan	Monitor	East	1103	2

The screenshot shows the same Jupyter Notebook interface. The code cell contains the following Python code:

```
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# 1. Rows where Region = "East"
print("Rows where Region = East:")
display(df[df['Region'] == 'East'])

# 2. Rows where Sales > 1500
print(["Rows where Sales > 1500:"])
display(df[df['Sales'] > 1500])
```

The output of the second code block is displayed as a table:

Order_ID	Customer	Product	Region	Sales	Quantity	
52	1053	Rahul	Laptop	East	2276	1
53	1054	Raj	Monitor	East	1420	2
64	1065	Amit	Monitor	East	815	8
66	1067	Rahul	Tablet	East	1875	6
71	1072	Anita	Keyboard	East	1702	3
76	1077	Vikas	Keyboard	East	978	6
77	1078	Vikas	Laptop	East	538	4
78	1079	Meera	Monitor	East	1295	7
79	1080	Pooja	Tablet	East	1829	2
81	1082	Anita	Phone	East	1695	7
82	1083	Neha	Keyboard	East	2034	5
85	1086	Pooja	Laptop	East	1330	2
87	1088	Karan	Monitor	East	875	2

Below the table, the text "Rows where Sales > 1500:" is displayed, followed by another table:

Order_ID	Customer	Product	Region	Sales	Quantity	
2	1003	Meera	Laptop	South	1597	5
4	1005	Priya	Phone	South	2028	4
9	1010	Amit	Phone	West	1998	4
10	1011	Raj	Monitor	North	1693	6
13	1014	Rohan	Phone	South	2078	3
16	1017	Neha	Laptop	East	2015	2
17	1018	Anita	Phone	North	1637	3
19	1020	Neha	Laptop	West	1503	6
22	1023	Rahul	Laptop	South	2267	3

Pandas DataFrames.ipynb

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Files

..

.config

sample_data

Retail_dataset.csv

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24	1025	Meera	Monitor	West	1926	6
26	1027	Vikas	Laptop	North	1864	8
27	1028	Vikas	Laptop	North	1551	3
29	1030	Rohan	Monitor	West	1932	5
31	1032	Amit	Laptop	East	2226	6
34	1035	Amit	Tablet	East	2104	3
35	1036	Rohan	Phone	East	2136	8
36	1037	Raj	Phone	East	1650	4
37	1038	Amit	Monitor	East	1567	8
38	1039	Meera	Keyboard	East	2455	3
42	1043	Meera	Keyboard	West	1515	6
43	1044	Vikas	Laptop	West	2248	5
44	1045	Priya	Phone	East	1875	4
49	1050	Karan	Monitor	West	2034	7
52	1053	Rahul	Laptop	East	2276	1
57	1058	Meera	Tablet	North	1666	8
58	1059	Karan	Tablet	West	2404	6
59	1060	Karan	Tablet	West	1651	2
60	1061	Vikas	Monitor	South	1837	8
61	1062	Meera	Keyboard	North	1688	4
63	1064	Pooja	Keyboard	West	2310	5
66	1067	Rahul	Tablet	East	1875	6
69	1070	Rahul	Tablet	West	2134	5
71	1072	Anita	Keyboard	East	1702	3

🔍 Commands + Code ▾ + Text | ▶ Run all ▾

24	1025	Meera	Monitor	West	1926	6
26	1027	Vikas	Laptop	North	1864	8
27	1028	Vikas	Laptop	North	1551	3
29	1030	Rohan	Monitor	West	1932	5
31	1032	Amit	Laptop	East	2226	6
34	1035	Amit	Tablet	East	2104	3
35	1036	Rohan	Phone	East	2136	8
36	1037	Raj	Phone	East	1650	4
37	1038	Amit	Monitor	East	1567	8
38	1039	Meera	Keyboard	East	2455	3
42	1043	Meera	Keyboard	West	1515	6
43	1044	Vikas	Laptop	West	2248	5
44	1045	Priya	Phone	East	1875	4
49	1050	Karan	Monitor	West	2034	7
52	1053	Rahul	Laptop	East	2276	1
57	1058	Meera	Tablet	North	1666	8
58	1059	Karan	Tablet	West	2404	6
59	1060	Karan	Tablet	West	1651	2
60	1061	Vikas	Monitor	South	1837	8
61	1062	Meera	Keyboard	North	1688	4
63	1064	Pooja	Keyboard	West	2310	5
66	1067	Rahul	Tablet	East	1875	6
69	1070	Rahul	Tablet	West	2134	5
71	1072	Anita	Keyboard	East	1702	3

The screenshot shows the JupyterLab interface for a notebook named 'Pandas DataFrames.ipynb'. The top bar includes the JupyterLab logo, the notebook name, and icons for star and cloud. The menu bar contains 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. The toolbar below the menu bar includes a search icon, 'Commands', '+ Code', '+ Text', and 'Run all'. On the left, a file explorer shows the current directory with files like '.config' and 'sample_data'. The main area displays a Pandas DataFrame with 7 rows and 7 columns.

74	1075	Amit	Monitor	North	2390	5
79	1080	Pooja	Tablet	East	1829	2
81	1082	Anita	Phone	East	1695	7
82	1083	Neha	Keyboard	East	2034	5
88	1089	Raj	Tablet	North	2213	5

🔍 Commands + Code ▾ + Text | ▶ Run all ▾

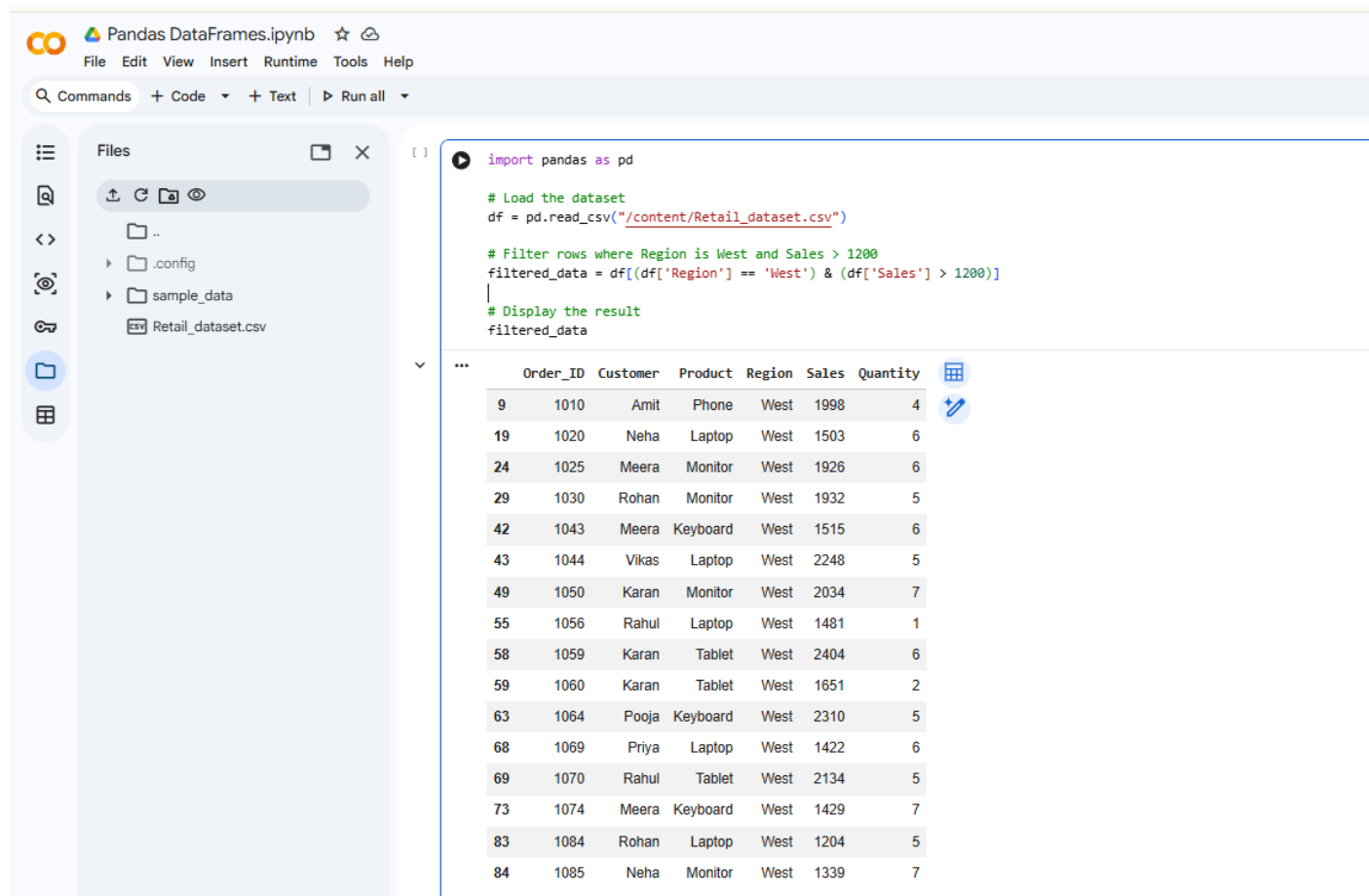
74	1075	Amit	Monitor	North	2390	5
79	1080	Pooja	Tablet	East	1829	2
81	1082	Anita	Phone	East	1695	7
82	1083	Neha	Keyboard	East	2034	5
88	1089	Raj	Tablet	North	2213	5

Q8. Multiple Condition Filtering

Display rows where:

1. Region is West
2. Sales is greater than 1200

Ans:-



The screenshot shows a Jupyter Notebook interface with the following components:

- Files Panel:** Displays a file explorer with the following structure:
 - ..
 - .config
 - sample_data
 - Retail_dataset.csv
- Code Cell:** Contains the following Python code:

```
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# Filter rows where Region is West and Sales > 1200
filtered_data = df[(df['Region'] == 'West') & (df['Sales'] > 1200)]

# Display the result
filtered_data
```
- Data Table:** Displays the filtered data as a table with 7 columns: Order_ID, Customer, Product, Region, Sales, Quantity. The table contains 16 rows of data, all from the 'West' region with sales greater than 1200.

	Order_ID	Customer	Product	Region	Sales	Quantity
9	1010	Amit	Phone	West	1998	4
19	1020	Neha	Laptop	West	1503	6
24	1025	Meera	Monitor	West	1926	6
29	1030	Rohan	Monitor	West	1932	5
42	1043	Meera	Keyboard	West	1515	6
43	1044	Vikas	Laptop	West	2248	5
49	1050	Karan	Monitor	West	2034	7
55	1056	Rahul	Laptop	West	1481	1
58	1059	Karan	Tablet	West	2404	6
59	1060	Karan	Tablet	West	1651	2
63	1064	Pooja	Keyboard	West	2310	5
68	1069	Priya	Laptop	West	1422	6
69	1070	Rahul	Tablet	West	2134	5
73	1074	Meera	Keyboard	West	1429	7
83	1084	Rohan	Laptop	West	1204	5
84	1085	Neha	Monitor	West	1339	7

Q9. Descriptive Statistics

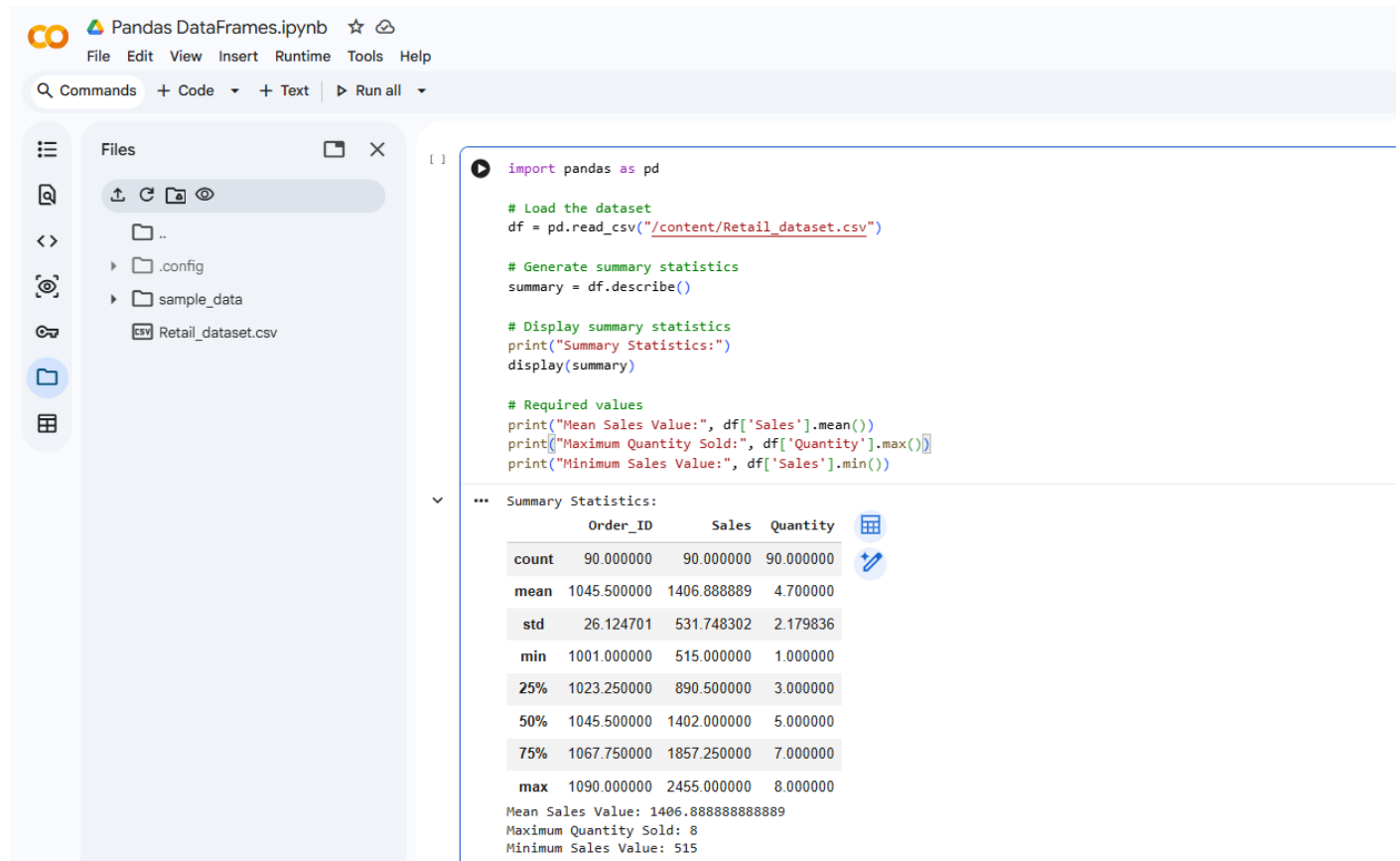
Generate summary statistics using:

describe()

From the result identify:

- Mean sales value
- Maximum quantity sold
- Minimum sales value

Ans:-



The screenshot shows a Jupyter Notebook interface with a file explorer on the left and a code editor on the right. The file explorer shows a directory structure with files like `..`, `.config`, `sample_data`, and `Retail_dataset.csv`. The code editor contains the following Python code:

```
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# Generate summary statistics
summary = df.describe()

# Display summary statistics
print("Summary Statistics:")
display(summary)

# Required values
print("Mean Sales Value:", df['Sales'].mean())
print("Maximum Quantity Sold:", df['Quantity'].max())
print("Minimum Sales Value:", df['Sales'].min())
```

The output of the code is a summary statistics table for the DataFrame:

	Order_ID	Sales	Quantity
count	90.000000	90.000000	90.000000
mean	1045.500000	1406.888889	4.700000
std	26.124701	531.748302	2.179836
min	1001.000000	515.000000	1.000000
25%	1023.250000	890.500000	3.000000
50%	1045.500000	1402.000000	5.000000
75%	1067.750000	1857.250000	7.000000
max	1090.000000	2455.000000	8.000000

Below the table, the following values are printed:

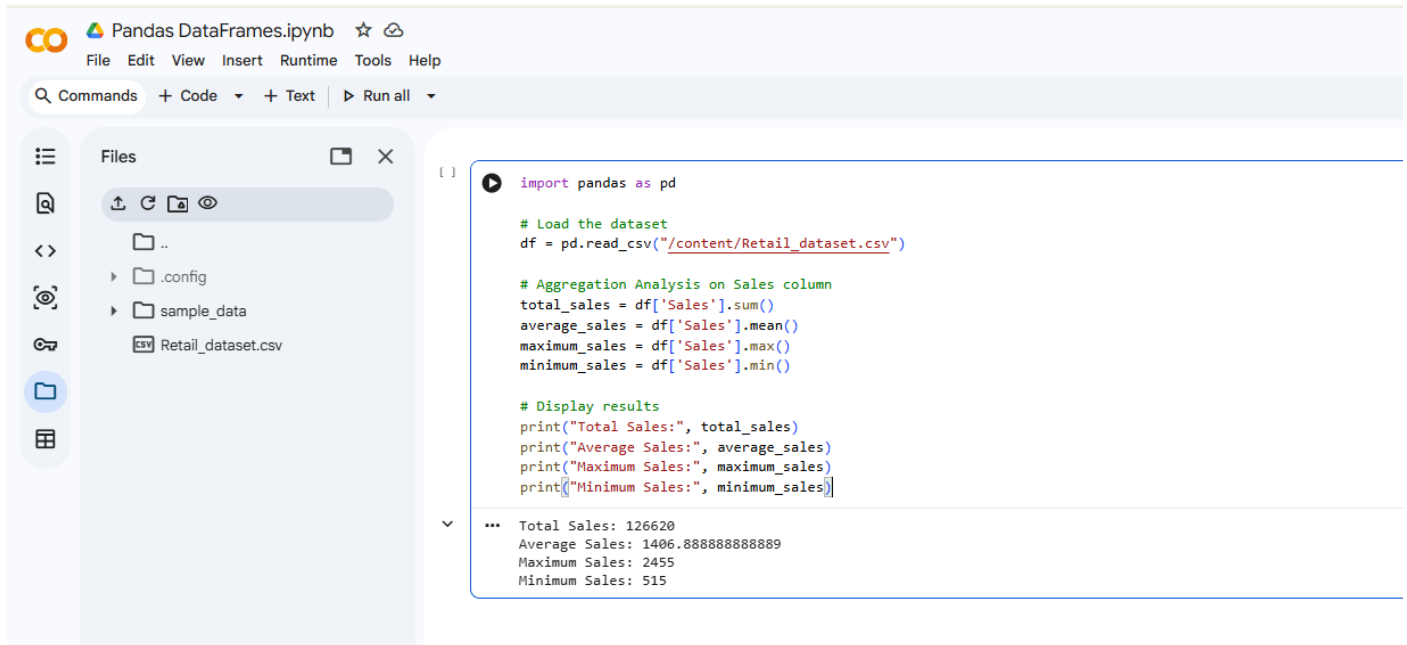
```
Mean Sales Value: 1406.888888888889
Maximum Quantity Sold: 8
Minimum Sales Value: 515
```

Q10. Aggregation Analysis

Using the Sales column, calculate:

1. Total sales
2. Average sales
3. Maximum sales
4. Minimum sales

Ans:-



The screenshot shows a Jupyter Notebook interface with the title 'Pandas DataFrames.ipynb'. The left sidebar displays a file explorer with the following structure:

- Files
 - ..
 - .config
 - sample_data
 - Retail_dataset.csv

The main code cell contains the following Python code:

```
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# Aggregation Analysis on Sales column
total_sales = df['Sales'].sum()
average_sales = df['Sales'].mean()
maximum_sales = df['Sales'].max()
minimum_sales = df['Sales'].min()

# Display results
print("Total Sales:", total_sales)
print("Average Sales:", average_sales)
print("Maximum Sales:", maximum_sales)
print("Minimum Sales:", minimum_sales)
```

The output of the code cell is displayed below the code:

```
*** Total Sales: 126620
Average Sales: 1406.888888888889
Maximum Sales: 2455
Minimum Sales: 515
```